Closed Loop Control of Single-phase Boost PFC Converter - III

Introduction

In today's energy-conscious world, efficient power delivery is paramount. Power Factor Correction (PFC) converters play a crucial role in ensuring that electronic devices draw current from the grid in a way that minimizes reactive power, thereby improving overall system efficiency and reducing harmonic distortion. However, achieving optimal performance from these converters requires sophisticated control strategies. This lecture delves into the intricate details of closed-loop control for single-phase boost PFC converters, building upon previous discussions. Understanding these control mechanisms is not just an academic exercise; it's fundamental to designing robust and reliable power systems that meet stringent regulatory standards and deliver stable power to critical loads. The focus here is on the mathematical derivation of the converter's dynamic behavior, which is essential for designing effective controllers that can regulate output voltage while maintaining a unity power factor.

Core Concepts

Term	Definition	Significance
`Closed-loop Control`	A control system where the output is measured and fed back to adjust the input, ensuring the system maintains desired performance despite disturbances.	Essential for regulating output voltage and achieving unity power factor in PFC converters, ensuring stability and efficiency.
`Unity Power Factor (UPF)`	A condition where the current drawn from the source is in phase with the voltage, minimizing reactive power and harmonic distortion.	A primary objective of PFC converters, crucial for grid compatibility, reduced energy losses, and compliance with power quality standards.
	A control strategy that regulates the average value	Enables the converter to draw a sinusoidal input

Term	Definition	Significance
`Average Current Control`	of the inductor current to shape the input current waveform.	current in phase with the input voltage, thereby achieving unity power factor.
`Gi(s)`	The transfer function representing the dynamic relationship between the inductor current (output) and the duty ratio (input) in the s-domain.	Crucial for designing the inner current loop controller, as it describes how the inductor current responds to changes in the control signal.
`Gv(s)`	The transfer function representing the dynamic relationship between the output voltage (output) and the inductor current (input) in the s-domain.	Essential for designing the outer voltage loop controller, as it describes how the output voltage responds to changes in the inductor current.
`Small-Signal Model`	A linearized model of a non-linear system, derived by considering small perturbations around a steady-state operating point.	Simplifies the analysis of complex converter dynamics, allowing for the application of linear control theory to design controllers.
`State-Space Representation`	A mathematical model of a physical system as a set of input, output, and state variables related by first-order differential equations.	Provides a comprehensive framework for analyzing and designing controllers for multi-input, multi-output systems like power converters.
`Perturbation`	A small, incremental change applied to a system variable around its steady-state operating point.	The basis for deriving small-signal models, allowing the linearization of non-linear system equations.
`Linearization`	The process of approximating a non-linear system with a linear model, typically by	Simplifies control design by enabling the use of well-established linear control

Term	Definition	Significance
	considering small deviations from an operating point.	techniques, such as transfer function analysis.
`Transfer Function`	A mathematical representation, in the s-domain, of the relationship between the output and input of a linear time-invariant system.	A fundamental tool in control theory for analyzing system stability, frequency response, and designing compensators.

Detailed Analysis

1. Control Objectives and Loop Structure

The primary goal of closed-loop control for a single-phase boost PFC converter is twofold:
- **Output Voltage Regulation**: The converter must maintain a stable and regulated DC output voltage, regardless of load variations or input voltage fluctuations.
- **Unity Power Factor (UPF) Current Drawing**: The converter must draw a sinusoidal input current that is in phase with the input voltage, ensuring high power quality and efficiency.

To achieve these objectives, a hierarchical control structure is employed, consisting of:
- **Outer Voltage Loop**: This loop is responsible for regulating the output DC voltage. It typically generates a reference signal for the inner current loop.
- **Inner Current Loop**: This loop ensures that the input current follows the reference generated by the outer loop, thereby achieving average current control and unity power factor. This involves controlling the average value of the inductor current to match a desired sinusoidal waveform.

2. Deriving Transfer Functions for Controller Design

To effectively design the controllers for both the inner and outer loops, it is crucial to understand the dynamic behavior of the converter. This understanding is captured through transfer functions:
- `Gi(s)` **(Current Loop Transfer Function)**: This represents the relationship between the inductor current `iL(s)` and the duty ratio `d(s)`. It describes how the inductor current responds to changes in the control signal (duty ratio).
- `Gv(s)` **(Voltage Loop Transfer Function)**: This represents the relationship

between the output voltage `Vo(s)` and the inductor current `iL(s)`. It describes how the output voltage responds to changes in the current flowing through the inductor.

The derivation of `Gi(s)` is the primary focus of this lecture, as it is fundamental to designing the inner current control loop.

3. Small-Signal Model Derivation: The Path to Linearization

The boost PFC converter is inherently a non-linear system due to the switching action and the product of variables (e.g., duty ratio and voltage). To apply linear control theory, a small-signal model is derived through linearization.

- **Average Large-Signal Model**: The starting point is the average large-signal model, typically represented in state-space form. This model describes the average behavior of the converter over a switching period using state variables like inductor current (`iL`) and capacitor voltage (`vc`), and input variables like input voltage (`Vs`) and duty ratio (`D`). The state-space matrices (A, B, C) are derived by averaging the converter's behavior during the `DTs` (switch ON) and `(1-D)Ts` (switch OFF) periods.
- **Introducing Small Perturbations**: To linearize the system, small perturbations are introduced around the steady-state operating point. This means each variable is expressed as its steady-state value plus a small, time-varying perturbation:
- Duty ratio: `D = D + d_tilde`
- Inductor current: `iL = iL + iL_tilde`
- Capacitor voltage: `vc = vc + vc_tilde`
- Input voltage: `Vs = Vs + Vs_tilde`
  The condition for these perturbations is that the tilde (~) terms are much smaller than their corresponding steady-state (capital) values.
- **Linearization through Approximation**: The key step in linearization is to assume that the product of two small perturbation terms is negligible and can be approximated as zero. For example, `d_tilde * x_tilde ≈ 0`. This approximation simplifies the non-linear equations into linear ones.
- **Separating Steady-State and Small-Signal Terms**: After substituting the perturbed variables into the average large-signal state-space equations, the equations are separated into two sets:
- **Steady-State Equations**: These describe the converter's behavior at its operating point, where all perturbation terms are zero. For steady-state, the derivatives of the state variables are also zero (`dx/dt = 0`). This allows for

the calculation of steady-state values for `X` (state vector, e.g., `iL`, `vc`) and `Vo`. For instance, `Vo = Vs / (1-D)` is a common steady-state relationship for a boost converter.

- **Small-Signal Equations**: These describe how the small perturbations interact and evolve over time. These are the linearized differential equations that will be used to derive the transfer functions.

4. Deriving the Small-Signal State Equations

The linearized small-signal state equations take the form:
`dx_tilde / dt = A * x_tilde + B * Vs_tilde + K * d_tilde`
`vo_tilde = C * x_tilde + Q * d_tilde`

Here:
* `x_tilde` is the vector of small-signal state variables (`iL_tilde`, `vc_tilde`).
* `Vs_tilde` is the small-signal input voltage perturbation.
* `d_tilde` is the small-signal duty ratio perturbation.
* `A` and `B` are the state-space matrices from the average model.
* `K` and `Q` are new matrices derived from the linearization process, capturing the influence of the duty ratio perturbation on the state variables and output, respectively. The `K` matrix, for example, incorporates terms like `(A1 - A2)X + (B1 - B2)Vs`.

5. From Time Domain to S-Domain: Laplace Transform

To obtain transfer functions, the linearized small-signal differential equations are transformed into the s-domain using the Laplace transform.
Applying the Laplace transform to the state equation yields:
`s * x_tilde(s) = A * x_tilde(s) + B * Vs_tilde(s) + K * d_tilde(s)`
Rearranging this to solve for `x_tilde(s)`:
`(sI - A) * x_tilde(s) = B * Vs_tilde(s) + K * d_tilde(s)`
`x_tilde(s) = (sI - A)^-1 * [B * Vs_tilde(s) + K * d_tilde(s)]`

Similarly, for the output equation:
`vo_tilde(s) = C * x_tilde(s) + Q * d_tilde(s)`

6. Calculating the Current Loop Transfer Function, `Gi(s)`

To find `Gi(s) = iL(s) / d(s)`, we assume that there is no perturbation in the input voltage, meaning `Vs_tilde(s) = 0`.
Substituting `Vs_tilde(s) = 0` into the `x_tilde(s)` equation:
`x_tilde(s) = (sI - A)^-1 * K * d_tilde(s)`

Since `x_tilde(s)` is a vector containing `iL_tilde(s)` and `vc_tilde(s)`, the `iL_tilde(s)` component can be isolated. The lecture details the matrix inversion of `(sI - A)` and subsequent multiplication by `K`.

The final derived form of `Gi(s)` (the transfer function from duty ratio to inductor current, assuming constant input voltage) is:
`Gi(s) = iL(s) / d(s) = [Vs / (RL * (1 - D)^3)] * [(RLC * S + 2) / (S^2 * LC / (1 - D)^2 + S * L / (RL * (1 - D)^2) + 1)]`

This transfer function reveals important dynamic characteristics:
* **Second-order denominator**: Indicates the presence of two poles, suggesting a resonant behavior.
* **First-order numerator**: Indicates a single zero.

These characteristics are crucial for designing a stable and responsive current controller.

7. Next Steps: `Gv(s)` and Controller Parameters

The derivation of `Gi(s)` is a significant step towards understanding the converter's dynamics. The next logical step, as indicated, would be to derive `Gv(s)`, which relates the output voltage to the inductor current. With both `Gi(s)` and `Gv(s)` in hand, engineers can then proceed to design the appropriate compensators for the inner current loop and the outer voltage loop, ultimately achieving the desired closed-loop control objectives of regulated output voltage and unity power factor current drawing from the AC source.

Key Takeaways

- [ ] Closed-loop control of PFC converters requires regulating output voltage and drawing unity power factor current.
- [ ] The control structure typically involves an outer voltage loop and an inner current loop.
- [ ] `Gi(s)` (inductor current to duty ratio) and `Gv(s)` (output voltage to inductor current) are essential transfer functions for controller design.
- [ ] Small-signal modeling linearizes the non-linear converter dynamics by considering small perturbations around an operating point.
- [ ] The linearization process involves approximating products of small perturbation terms as zero.
- [ ] Steady-state and small-signal equations are separated from the perturbed average large-signal model.
- [ ] The Laplace transform is applied to the linearized small-signal equations to obtain transfer functions in the s-domain.

- [ ] `Gi(s)` for the boost PFC converter is a second-order system with a single zero, indicating its dynamic response characteristics.

Conclusion

The detailed derivation of the small-signal model and the `Gi(s)` transfer function provides the mathematical blueprint necessary for designing effective closed-loop controllers for single-phase boost PFC converters. By understanding how the inductor current dynamically responds to changes in the duty ratio, engineers can precisely tune the inner current loop to ensure the converter draws a near-perfect sinusoidal current from the grid, achieving unity power factor. This rigorous analytical approach, moving from large-signal averaging to small-signal linearization and finally to s-domain transfer functions, is a cornerstone of power electronics control design. It enables the creation of robust, stable, and efficient power conversion systems that are critical for modern electrical infrastructure and a wide array of electronic applications. The insights gained from this derivation directly inform the selection of controller parameters, ensuring optimal performance in real-world scenarios.